

AI HACKATHON · SMARTER PATIENT CARE

AI for Smarter Patient Care

One unified patient timeline. Answers grounded only in evidence you can trace back to the source.

Research & educational prototype only — not for clinical use. Do not use for diagnosis, treatment, triage, or emergency decisions.

Live demo: aiforsmarterpatientcare.streamlit.app

Dataset: MIMIC-IV Clinical Database Demo v2.2 · 100 patients · 149,673 unified clinical events

THE PROBLEM

One patient's story, scattered across dozens of tables

1

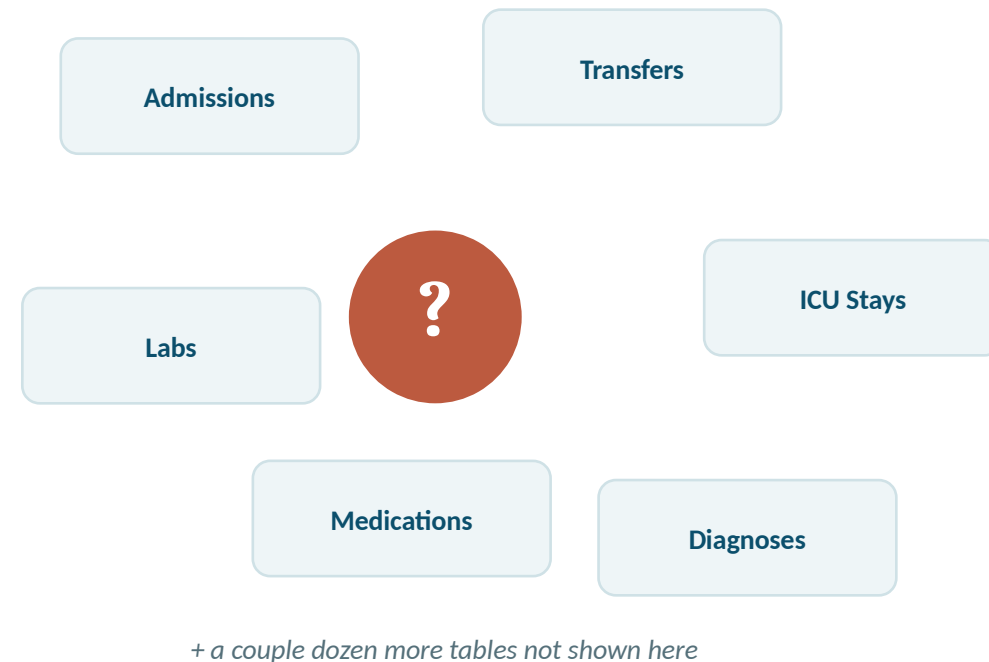
MIMIC-IV's schema has a couple dozen linked tables. Admissions, transfers, labs, medications, diagnoses, and ICU readings each live in their own place — and that's before procedures, prescriptions, or microbiology results.

2

Answering something as simple as “what was the highest creatinine reading” means manually joining several tables by hand and reconciling timestamps.

3

Most patients here were admitted more than once — mix up which visit you're looking at, and two unrelated hospital stays can read as one continuous story.



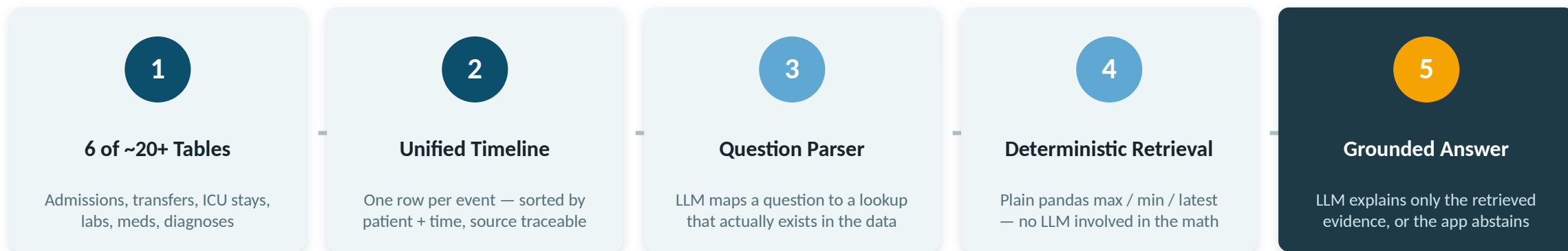
100 patients across **275** admissions

Most patients were admitted more than once.

OUR SOLUTION

Evidence first. The AI explains — it never invents.

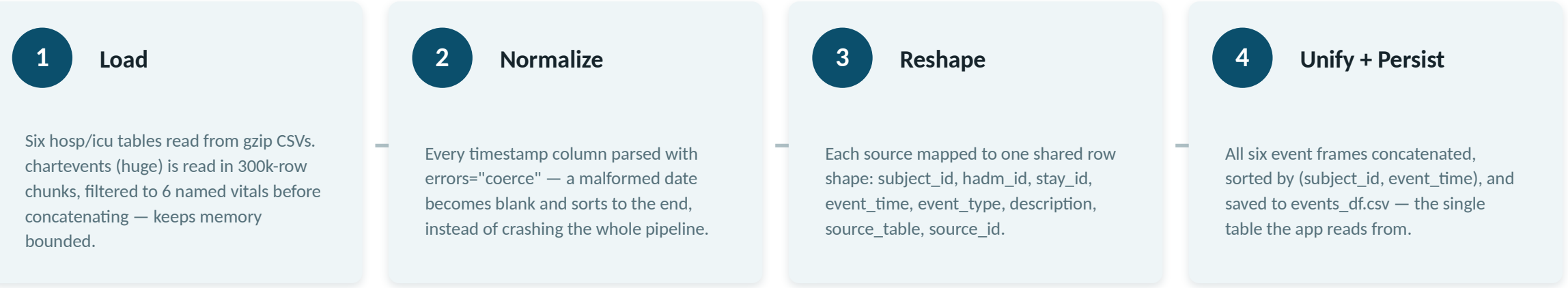
We deliberately scoped to the six tables a clinical timeline actually needs — visit history, labs, meds, and diagnoses. Procedures, prescriptions, and microbiology results add depth later; they weren't required for this track's core evidence-retrieval loop.



The patient data is the source of truth — not the model.

If no evidence is found, the app abstains with a specific reason instead of guessing. If evidence is found, the model is instructed to state only what's in it, and to cite the source table and timestamp.

From raw MIMIC-IV tables to one unified timeline



Four generated files, one for each thing the app needs

events_df.csv	lab_qa_flagged.csv	patient_info.csv	vitals_df.csv
The unified timeline — every event, every patient	Lab results plus data-quality flags	Demographics + first-admission facts	Six key ICU vital signs, filtered from chartevents

Two functions do all of the real answering

No LLM call happens until one of these two functions has already run. Both return the same shape — status, evidence, and a reason if nothing qualifies — so the UI treats every kind of question identically.

`retrieve_lab_metric(subject_id, metric, agg)`

- Filters lab_qa_flagged to this patient and this exact metric label
- Keeps only rows with a numeric value
- `agg="max" / "min"` → `idxmax / idxmin`
- `agg="latest"` → sorted by charttime, last row
- Abstains if the metric is unrecognized, has no numeric records, or every record is flagged implausible

`retrieve_event(subject_id, event_type)`

- Filters events_df to this patient and this exact event type
- Sorted chronologically by event_time
- Returns every matching row as evidence (up to 10)
- Abstains if the event type is unrecognized
- Abstains if no matching records exist for this patient

When the app says “I don't know” — and why

Not the model's judgment call — five hard rules that fire before it ever gets a second chance to speak.

1

The requested metric or event type isn't in the data at all

2

No numeric records exist for that lab metric for this patient

3

Every matching lab record is flagged implausible

4

No matching event records exist for that event type, for this patient

5

The question doesn't map to a supported lookup type

Every abstention comes with a specific reason — never a bare “no.”

Every lab result gets checked. Nothing gets changed.

What this table shows: the four independent checks a lab row can be flagged with — a row can carry more than one. Flags are additive metadata on a copy of the data; the original values are never modified or deleted.

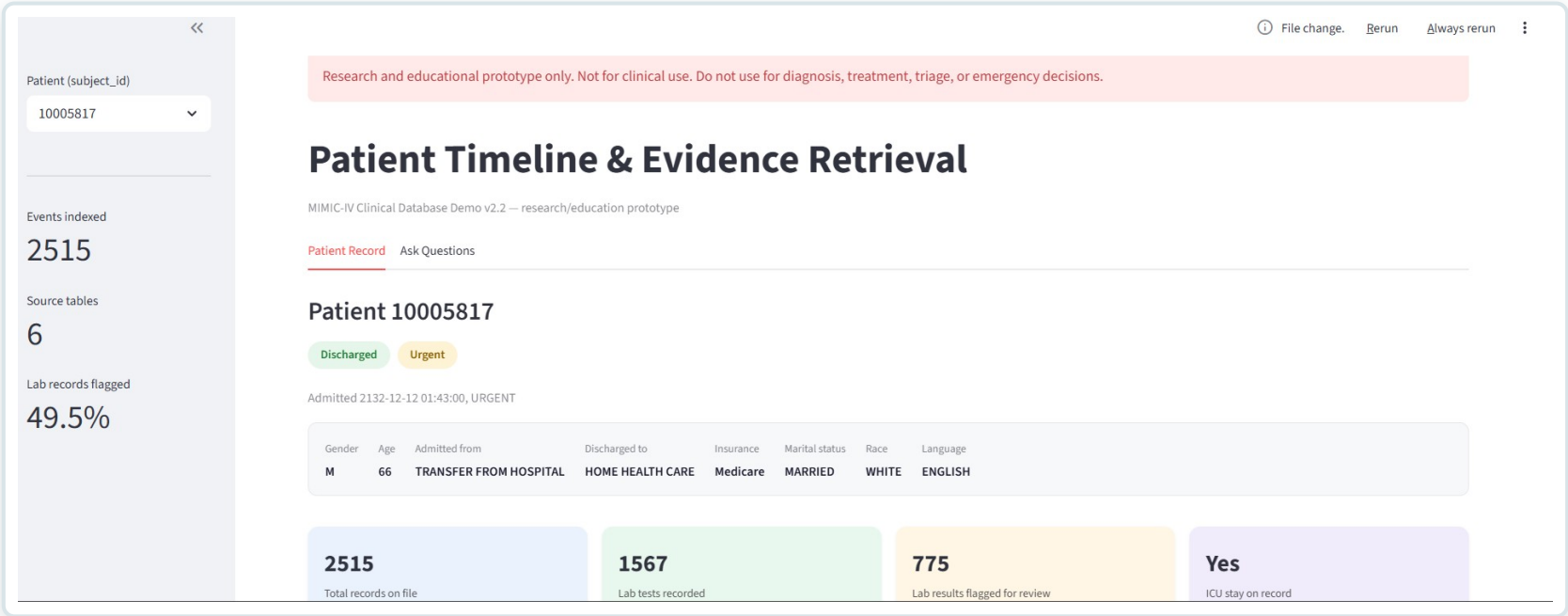
Flag	What triggers it
missing_unit	Value is present but the unit field is blank
lab_marked_abnormal	Source flag field equals "abnormal"
duplicate_result	Same patient + test + time + value appears more than once
implausible_value	Value falls outside a plausible range for 7 named tests (e.g. Creatinine, Hemoglobin, Glucose)

Flagged doesn't mean incorrect — it means a human should look before relying on it.

Flagged records are shown as-is in a dedicated view, never hidden, corrected, or silently dropped.

THE PRODUCT

Every patient's record, unified on one screen



Live screenshot — aiforsmarterpatientcare.streamlit.app



Status at a glance

Discharged / Admitted / In ICU, plus admission type — read in two seconds.



KPI cards

Total records, lab tests, flagged results, and ICU stay — all computed live from the data.



Demographics strip

Age, gender, insurance, and disposition, pulled from the patient_info table.

THE PRODUCT

Six central tables, merged into one traceable timeline

Full timeline

What this shows: every event for this patient, admissions, transfers, labs, medications, diagnoses, and ICU stays, merged from 6 separate RDMC HIE tables into one time ordered list. The last two columns show exactly which source table and row each event came from, so any entry can be traced back and verified.

Time	Type	Description	Source Table	Source Row ID
236 2180-05-07 05:05:00	Laboratory	Platelet Count: 71 K/uL	labevents	138
337 2180-05-07 05:05:00	Laboratory	RDW: 15.3 %	labevents	139
339 2180-05-07 05:05:00	Laboratory	PTT: 32.3 sec	labevents	132
340 2180-05-07 05:05:00	Laboratory	Bilirubin, Total: 1.3 mg/dL	labevents	147
351 2180-05-07 05:05:00	Laboratory	PT: 36.6 sec	labevents	131
355 2180-05-07 05:05:00	Laboratory	MCHC: 33.6 %	labevents	136
360 2180-05-07 05:05:00	Laboratory	INR/PTT: 1.5	labevents	130
347 2180-05-07 05:05:00	Laboratory	Bicarbonate: 28 mEq/L	labevents	146
346 2180-05-07 05:05:00	Laboratory	Potassium: 4.5 mEq/L	labevents	154
335 2180-05-07 05:05:00	Laboratory	Glucose ____ mg/dL	labevents	151

Flagged lab records

What this shows: lab results that failed one of four automatic quality checks (missing unit, marked abnormal by the lab, duplicate entry, or a value outside a plausible range). Nothing here has been changed or deleted; these are shown as-is, flagged for a human reviewer to judge.

Time	Test	Value	Unit	Why it's flagged
3338 2180-08-10 12:00:00	Sodium	124	mEq/L	lab_marked_abnormal
3339 2180-08-10 12:00:00	Red Blood Cells	3.27	m/dL	lab_marked_abnormal
3337 2180-08-10 12:00:00	MCH	83	%	lab_marked_abnormal
3339 2180-08-10 12:00:00	Neutrophils	76.8	%	lab_marked_abnormal
3340 2180-08-10 12:00:00	Lymphocytes	12.8	%	lab_marked_abnormal
3342 2180-08-10 12:00:00	Hematscrit	33.7	%	lab_marked_abnormal
3343 2180-08-10 12:00:00	Platelet Count	136	K/uL	lab_marked_abnormal
3344 2180-08-10 12:00:00	Hemoglobin	11.6	g/dL	lab_marked_abnormal
3345 2180-08-10 12:00:00	RDW	16.1	%	lab_marked_abnormal
3346 2180-08-10 12:00:00	MCH	85.3	pg	lab_marked_abnormal

Every row keeps its source table and row ID — nothing here is a black box.

Latest Lab Results

What this shows: the most recent recorded value for each lab test this patient has had, one row per test, sorted by date. Rows marked 'Flagged' were caught by an automatic data quality check (missing unit, abnormal per the lab's own reference range, duplicate entry, or an implausible number). Flagged does not mean incorrect; it means it needs a human look before it's OK.

Test	Value	Unit	Recorded	Status
Sodium	124	mEq/L	2180-08-10 12:00:00	Abnormal
Glucose	155	mg/dL	2180-08-10 12:00:00	Abnormal
MCH	85.3	pg	2180-08-10 12:00:00	Abnormal
Hematscrit	33.7	%	2180-08-10 12:00:00	Abnormal
RDW	16.1	%	2180-08-10 12:00:00	Abnormal
Hemoglobin	11.6	g/dL	2180-08-10 12:00:00	Abnormal
Platelet Count	136	K/uL	2180-08-10 12:00:00	Abnormal
Monocytes	9.7	%	2180-08-10 12:00:00	Normal
Basophils	0.2	%	2180-08-10 12:00:00	Normal
Neutrophils	76.8	%	2180-08-10 12:00:00	Abnormal

Latest ICU Vitals

What this shows: the most recent bedside reading for each core vital sign, from the ICU chartevents table. Only present if this patient had an ICU stay with recorded observations.

Vital Sign	Value	Unit	Recorded
Non Invasive Blood Pressure diastolic	85	mmHg	2180-07-23 22:00:00
Non Invasive Blood Pressure systolic	85	mmHg	2180-07-23 22:00:00
Respiratory Rate	20	breaths/min	
O2 saturation pulseoxymetry	95	%	2180-07-23 22:00:00
Heart Rate	94	bpm	2180-07-23 22:00:00
Temperature Fahrenheit	98.5	°F	2180-07-23 20:00:00

Lab results carry a data-quality status — *flagged*, not hidden or auto-corrected.

AI for Smarter Patient Care — Research & educational prototype. Not for clinical use.

Track 1 — Structured Patient Timeline & Evidence Retrieval

Ask a real question. Get a grounded answer.

Patient (subject_id)

10000032

Events indexed

821

Source tables

6

Lab records flagged

47.4%

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Patient Timeline & Evidence Retrieval

MIMIC-IV Clinical Database Demo v2.2 — research/education prototype

Patient RecordAsk Questions

Patient 10000032

Ask a structured-data question

e.g. What was the highest creatinine value?

What was the highest creatinine value?

The highest creatinine value was 0.8 mg/dL (source_table: labevents, timestamp: 2180-08-10 12:00:00).

Show parsed query + evidence

Full timeline

What this shows: every event for this patient, admissions, transfers, labs, medications, diagnoses, and ICU stays, merged from 6 separate MIMIC-IV tables into one time ordered list. The last two columns show exactly which source table and row each event came from, so any entry can be traced back and verified.

	Time	Type	Description	Source Table	Source Row ID
0	2180-03-23 11:51:00	Laboratory	Length of Urine Collection: Not available	labevents	56
55	2180-03-23 11:51:00	Laboratory	Sodium: 136 mEq/L	labevents	34
54	2180-03-23 11:51:00	Laboratory	Triglycerides: ____ mg/dL	labevents	38
53	2180-03-23 11:51:00	Laboratory	Cholesterol Ratio (Total/HDL): 3.9 Ratio	labevents	15
52	2180-03-23 11:51:00	Laboratory	Alpha-Fetoprotein: ____ ng/mL	labevents	7
51	2180-03-23 11:51:00	Laboratory	Calcium, Total: 8.4 mg/dL	labevents	12

“What was the highest creatinine value?”

The highest creatinine value was 0.8 mg/dL (source_table: labevents, timestamp: 2180-08-10 12:00:00).

Every number in that sentence is cited back to its source row — not generated from the model's own knowledge.

Captured live from the deployed app.

Fixing the exact failure that already happened live

The app hit a real Groq rate limit in production before this review. Here's the before and after — no change to prompts, models, or answers, only to what happens when the API says no.



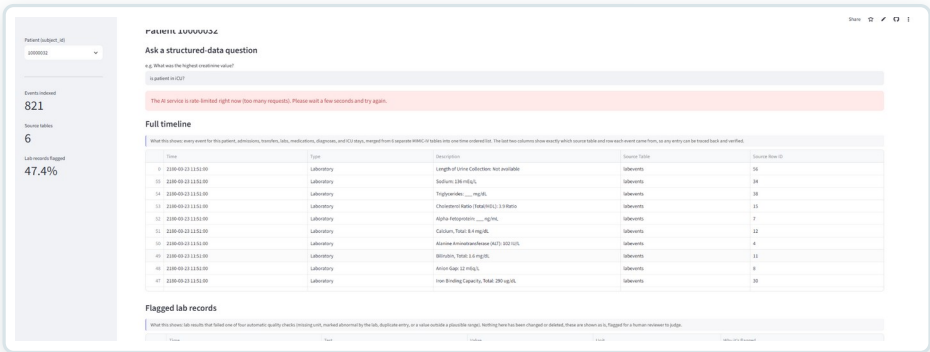
Before

- No error handling around either Groq call
- A 429 or timeout → raw Streamlit traceback
- Repeated reruns could re-fire the same two LLM calls for an unchanged question



After

- Retry with exponential backoff (2 retries) on 429s, timeouts, and 5xx errors
- Friendly in-app message on final failure — no raw traceback
- Answers cached per (patient, question) for the session — no redundant calls



Confirmed live on the deployed app — not just in code.

What we actually verified

What this table shows: seven representative questions we ran against the app or traced through the code, the behavior each should trigger, and how we confirmed it — manual, scenario-based testing, not a quantitative benchmark. No accuracy or hallucination-rate numbers are claimed, because none were measured.

Test case	Expected behavior	Result
“What was the highest creatinine value?”	Deterministic max, evidence cited	✓ Confirmed live
“What medications were given?”	Answered if EMAR exists, else abstains	✓ Confirmed
“What is the weather today?”	Abstains — unsupported question type	✓ Confirmed
Lab metric that doesn't exist	Abstains — names the exact problem	✓ Confirmed
Metric with zero numeric records	Abstains with a stated reason	✓ Confirmed
All matching records flagged implausible	Abstains, evidence still shown	✓ Confirmed
Groq API rate-limited mid-question	Friendly message, not a crash	✓ Confirmed live

What we found reviewing our own code

Part of a hackathon submission judges trust is showing you looked hard for problems, not just showcasing what works.



Record-count column renders blank live

The "Record types for this patient" table shows event types but no counts, on the deployed app. Likely a pandas version difference — newer pandas changed how `value_counts().reset_index()` names columns. One-line fix identified, not yet applied.



Missing CSVs aren't handled gracefully

If the four generated data files aren't present, the app raises a plain `FileNotFoundError` instead of a guided message. Known gap.



One unused, broken notebook cell

An early exploratory cell has a missing bracket (syntax error). It's never executed by anything downstream, but should be deleted before submission for a clean read-through.

Built with explicit boundaries



Intended use

Researchers, educators, and clinical-data teams exploring structured MIMIC-IV data — not clinicians making patient-care decisions.



Not intended for

Diagnosis, treatment, triage, or emergency decisions, for any patient, real or synthetic. Never a substitute for clinical judgment.



Privacy

Data was de-identified upstream by PhysioNet. No free-text clinical notes anywhere. Only the evidence needed to answer one question leaves the app.



Human oversight

Every AI sentence ships with its evidence attached. Flagged lab records are shown, never silently corrected or hidden.

Where this actually breaks: admission mixing

We're not going to hide this one — it's the most significant gap we know about.



What the design intends

Every unified event already carries an admission ID (hadm_id). The plan was for the app to scope a timeline to one hospital visit at a time — so two unrelated admissions are never read as one continuous story.



What the shipped app does today

The dashboard and timeline currently scope only by patient, not by admission. Since most of the 100 patients have more than one of the dataset's 275 admissions, their separate hospital stays are shown merged into a single timeline.

Why we're shipping it anyway:

A rushed, untested scoping change this late felt riskier than a clearly documented limitation. It's the top item on our next-iteration list — not a surprise we're hoping no one asks about.

Technology stack and dataset

Technology stack

- **Python + pandas**
Data loading, normalization, the unified timeline
- **Streamlit**
The web application (app.py)
- **Groq API — llama-3.3-70b-versatile**
Question parsing (forced tool call) + grounded synthesis
- **python-dotenv**
Local environment variable loading
- **Jupyter Notebook**
Pipeline development and processing

No vector database, no embeddings, no RAG framework — retrieval is deterministic
pandas filtering, not similarity search.

Dataset — MIMIC-IV Demo v2.2

Patients	100
Total unified events	149,673
Laboratory events	107,727
Medication events	35,835
Diagnosis events	4,506
Transfer events	1,190
Admission events	275
ICU stay events	140

SUBMISSION

All six required deliverables, accounted for

✓	Working prototype	Live end-to-end demo on the supplied data	<i>aiforsmarterpatientcare.streamlit.app</i>
✓	Source code + run instructions	Reproducible, with dependencies and setup steps	<i>app.py, requirements.txt, README.md</i>
✓	Technical summary	Target user, data flow, AI method, assumptions	<i>README.md, ARCHITECTURE.md</i>
✓	Evaluation report	Baseline, test protocol, results, known limitations	<i>EVALUATION.md</i>
✓	Safety & data statement	Intended use, privacy, failure modes, human review	<i>RESPONSIBLE_AI.md</i>
✓	Demo & pitch	Problem, product, evidence, one honest failure case	<i>This deck</i>

WHY THIS MATTERS

A traceable answer in seconds — not a black box.

This is the pattern for trustworthy AI over structured health data: retrieve real evidence first, let the model interpret and explain it, never let it invent facts, and always show the work behind every answer.

100

patients

149,673

unified events

6

tables unified, of ~20+

Try it live: aiforsmarterpatientcare.streamlit.app

Built for researchers, educators, and clinical-data teams — not as a replacement for a clinician's judgment.